

Mason Lee

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EDUCATION

Michigan State University, College of Engineering | GPA: 3.71/4.0

Bachelor of Science, Computational Data Science

May 2026

- Honors College Member, Dean's List Fall 2023, Spring 2024, Fall 2024, Spring 2025
- Relevant Completed Classwork: Computational Modeling, Differential Equations, Matrix Algebra, Data Science, Probability & Statistics, Programming, Parallel Computing, Algorithms and Data Structures

EXPERIENCE & PROJECTS

Satellite-Based Wildfire Nowcasting Pipeline

- Designed an end-to-end wildfire nowcasting pipeline that ingests NASA FIRMS/VIIRS data, aggregates detections into ~2 km cell-day features, and outputs calibrated next-day ignition probabilities for any region
- Built a self-serve Streamlit app with bbox/date/source controls that orchestrates the full pipeline on demand, handles secrets, and renders an interactive Folium/Leaflet time-slider map with downloadable risk outputs
- Engineered a batched, cache-first HLS downloader using earthaccess; produced cloud-masked NDVI/NBR with rioxarray/rasterio and deterministic daily mosaics, eliminating redundant I/O and making runs reproducible
- Validated the model using spatial cross-validation and rare-event metrics such as AUC-PR and Brier score, calibrated probabilities for actionable thresholds, and added SHAP explanations to surface key drivers like vegetation indices and recent detections.

MSU Undergraduate Research, Lansing, MI

Student Researcher

June 2024 - Present

- Designed and implemented KNet and RealNVP neural network models in PyTorch for probability distribution estimation and generative modeling, with applications to high-dimensional scientific data.
- Experimented with model stability and training dynamics, applying techniques such as coupling layers, normalization, and loss function tuning to improve performance.
- Developed simulation pipelines for multi-agent systems, using nonparametric methods to infer interaction kernels from trajectory data.
- Strengthened research skills by translating theoretical algorithms from scientific literature into practical code implementations and communicating findings through discussions with faculty mentors.

MSU DSS Tech Team, East Lansing, MI

Student Info Tech

September 2023 - Present

- Troubleshoot and resolved hardware, software, and network issues across campus, improving staff productivity and minimizing downtime.
- Managed and closed support tickets through TeamDynamix, providing timely technical assistance via Outlook, Teams, phone, and in-person communication.
- Led regular maintenance and upgrade projects, including the replacement of three generations of systems, ensuring staff access to secure and up-to-date technology.

INVOLVEMENT

MSU SASE, East Lansing, MI

President & Conference Chair

April 2025 - Present

- Elected President to lead chapter operations, oversee the executive board, and represent 40+ members at university, regional, and national levels.
- Appointed Conference Chair for the 2025 Midwest Regional Conference, responsible for planning workshops, networking sessions, and keynote speakers for 200+ expected attendees.
- Coordinate cross-functional teams to execute professional development programs, community initiatives, and large-scale events that enhance member engagement and visibility.

SKILLS

Languages: Python, C/C++, BASH, R, SQL, HTML, CSS

Libraries: numpy, pandas, scikit-Learn, matplotlib, pytorch, folium, xgboost, rioxarray, rasterio, earthaccess

Software Proficiency: Jupyter Notebook, Visual Studio Code, PyCharm, Unix Shell, Vim, RStudio, Streamlit